

TOMAKOMAI, Japan

WMO: 474240

Lat: 42.6233N

Lon: 141.5472E

Elev: 26

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	13.0	1.6	6.0	14.4	4.1	6.8	17.1	24.8	36.8	22.4	35.3	4.4	350	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	78.5	71.1	76.1	69.9	74.0	69.0	73.0	76.2	71.6	74.3	70.3	73.0	8.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	118.0	75.2	70.6	112.9	73.7	69.2	107.5	72.3	36.7	76.3	35.5	74.5	34.4	73.2	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	18.0	15.5	5.2	83.8	3.7	2.6	2.5	85.6	0.3	87.1	-1.7	88.6	-4.4	90.5
			4.3	74.8	3.6	1.9	1.7	76.2	-0.4	77.3	-2.4	78.4	-5.1	79.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	24.9	25.6	32.5	41.4	50.4	57.4	65.2	69.3	64.1	52.2	40.3	29.6	(6)
(7)		DBStd	16.10	5.28	5.49	5.18	4.66	4.58	4.40	4.20	3.98	4.99	5.76	6.64	6.02	(7)
(8)		HDD50	3293	777	684	543	261	51	2	0	0	44	297	634		(8)
(9)		HDD65	7116	1242	1104	1008	708	454	230	50	8	76	396	741	1099	(9)
(10)		CDD50	1904	0	0	0	3	63	225	472	600	422	113	6	0	(10)
(11)		CDD65	250	0	0	0	0	0	3	56	143	48	0	0	0	(11)
(12)		CDH74	478	0	0	0	0	1	12	72	311	82	0	0	0	(12)
(13)		CDH80	36	0	0	0	0	0	1	3	30	2	0	0	0	(13)
(14)	Wind	WSAvg	6.8	7.0	7.0	7.4	7.2	6.6	6.1	5.9	6.4	7.2	7.3	7.2	6.8	(14)
(15)	Precipitation	PrecAvg	48.5	2.1	1.9	2.1	3.1	4.8	4.0	6.3	7.0	6.9	4.0	3.5	2.8	(15)
(16)		PrecMax	62.0	3.2	5.2	4.4	5.7	8.5	11.4	13.5	13.4	12.8	6.0	8.3	6.7	(16)
(17)		PrecMin	37.9	1.1	0.6	0.9	0.2	1.4	1.3	2.8	2.3	1.3	1.4	1.6	1.6	(17)
(18)		PrecStd	5.7	0.6	1.0	0.8	1.5	1.8	2.0	2.5	3.1	2.8	1.3	1.5	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	42.2	50.4	59.9	68.7	75.1	79.4	83.4	79.2	70.0	60.0	50.7	(19)
(20)		0.4%	MCWB	38.2	37.9	41.9	47.8	54.3	63.0	70.1	73.3	70.5	60.9	54.5	47.8	(20)
(21)		2%	DB	38.7	39.3	45.4	54.6	63.8	68.7	75.9	79.9	76.3	66.7	57.7	47.0	(21)
(22)		2%	MCWB	35.7	35.4	39.0	45.4	53.3	61.9	69.2	72.4	69.2	59.8	53.2	43.2	(22)
(23)		5%	DB	36.2	37.3	42.9	51.6	60.2	66.1	73.4	77.2	74.1	64.6	55.3	43.2	(23)
(24)		5%	MCWB	32.8	33.8	37.8	44.1	52.5	60.9	68.5	71.2	67.8	58.6	50.9	39.1	(24)
(25)		10%	DB	34.1	35.5	40.8	49.1	57.4	63.7	71.2	75.1	72.2	62.6	52.7	39.6	(25)
(26)		10%	MCWB	30.6	32.2	36.7	43.1	51.6	59.5	67.3	70.2	67.0	56.9	48.3	35.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	39.1	43.1	50.5	57.7	65.5	72.2	75.5	74.0	64.5	56.7	48.5	(27)
(28)		0.4%	MCDWB	41.3	41.0	47.0	56.3	63.5	70.6	76.4	80.9	76.5	67.0	58.8	50.3	(28)
(29)		2%	WB	36.1	36.3	40.7	47.8	55.6	63.1	70.6	73.8	72.0	61.8	54.2	43.7	(29)
(30)		2%	MCDWB	38.4	38.3	43.8	52.1	60.3	67.1	73.9	77.7	74.2	65.0	56.8	46.5	(30)
(31)		5%	WB	33.5	34.5	39.0	45.9	54.0	61.6	69.3	72.6	70.0	60.0	51.6	39.4	(31)
(32)		5%	MCDWB	35.6	36.7	41.9	49.7	58.1	65.0	72.1	75.7	72.7	63.6	54.8	42.6	(32)
(33)		10%	WB	31.0	32.5	37.3	44.2	52.6	60.1	68.0	71.4	68.2	58.0	48.7	35.8	(33)
(34)		10%	MCDWB	33.8	35.2	40.2	47.8	56.3	63.0	70.6	74.0	71.2	61.7	52.3	39.1	(34)
(35)	Mean Daily Temperature Range		MDBR	14.1	14.7	13.1	13.1	11.7	8.9	7.7	8.5	12.8	16.3	14.8	13.5	(35)
(36)		5% DB	MCDBR	14.3	14.5	15.1	17.8	17.4	13.4	11.9	12.3	14.0	16.7	18.0	16.4	(36)
(37)			MCWBR	12.6	12.1	11.1	10.9	9.7	7.7	6.8	6.6	8.8	11.7	15.1	14.5	(37)
(38)			5% WB	MCDWB	14.5	14.0	14.2	15.0	13.9	11.2	9.5	9.7	9.6	14.3	16.5	16.1
(39)		MCWBR		13.7	12.7	11.6	10.3	8.9	7.0	5.8	5.8	7.2	11.8	15.5	15.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.346	0.398	0.474	0.535	0.555	0.532	0.519	0.487	0.426	0.412	0.393	0.362	(40)	
(41)		taud	2.061	1.916	1.803	1.769	1.819	1.953	2.059	2.165	2.288	2.184	2.134	2.064	(41)	
(42)		Ebn at Noon	237	242	238	235	235	241	243	246	254	239	220	217	(42)	
(43)		Edh at Noon	35	48	61	68	66	58	51	45	37	36	33	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	509	769	1134	1462	1618	1553	1391	1334	1172	876	539	410	(44)	
(45)		RadStd	30	52	71	149	144	125	152	104	109	47	43	33	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page